Machine Learning Interview Questions

Machine Learning Fundamentals

1. What is Machine Learning, and how does it differ from traditional programming?
2. What are the main types of Machine Learning?
3. What is the difference between Artificial Intelligence, Machine Learning and Deep Learning?
4. What is the difference between supervised, unsupervised and reinforcement learning?
5. What is the difference between classification and regression?
6. What is the difference between training, validation and test datasets?
7. What are features and target variables in a Machine Learning dataset?
8. What is the difference between a model, an algorithm and a parameter?
9. What are the main stages of a typical Machine Learning project lifecycle?
10. What are the common challenges faced when building Machine Learning models?

Data Preprocessing and Feature Engineering

1. Why is data preprocessing important before training a Machine Learning model?
2. How do you identify and handle missing values in a dataset?
3. What is the difference between `SimpleImputer` and `KNNImputer` in Scikit-learn?
4. What is the difference between label encoding and one-hot encoding?
5. What is feature scaling, and why is it important?
6. What is the difference between normalisation and standardisation?
7. What is an outlier, and how can you detect and handle outliers?
8. What is feature engineering? Give practical examples.
9. What is feature selection, and how does it differ from feature extraction?
10. What is data leakage, and how can you prevent it during preprocessing?

Mathematics and Statistics for ML

1. What are mean, median and mode, and when is each useful?
2. What is variance, and how does it differ from standard deviation?
3. What is the difference between covariance and correlation?
4. What is the difference between Pearson and Spearman correlation?
5. What is a probability distribution? Explain normal, binomial and Bernoulli distributions.
6. What is Bayes' theorem, and how is it used in Machine Learning?
7. What are the null and alternative hypotheses in statistical hypothesis testing?
8. What is a p-value, and how should it be interpreted?
9. What is gradient descent, and how does it minimise a loss function?
10. What are vectors, matrices and matrix multiplication, and why are they important in ML?

Linear and Logistic Regression

1. What is Linear Regression, and what assumptions does it make?
2. What is the difference between simple and multiple Linear Regression?
3. What is the mathematical equation of a Linear Regression model?
4. What is the cost function used in Linear Regression?
5. What is Mean Squared Error (MSE), and why is it commonly used?
6. What is the difference between Linear Regression and Logistic Regression?
7. How does Logistic Regression use the sigmoid function to predict probabilities?
8. What is the decision threshold in Logistic Regression, and how does changing it affect predictions?
9. What is multicollinearity, and how can you detect and handle it?
10. What are the limitations of Linear Regression and Logistic Regression?

Classification Algorithms

1. What is a classification algorithm, and what are its common applications?
2. How does the K-Nearest Neighbours (KNN) algorithm work?
3. How does the choice of K affect the performance of KNN?

4. Why is feature scaling important for KNN?
5. How does a Support Vector Machine (SVM) classify data?
6. What is the difference between a linear SVM and a non-linear SVM?
7. What is the kernel trick in SVM? Explain linear, polynomial and RBF kernels.
8. How does Naive Bayes work, and why is it called "naive"?
9. What is the difference between Gaussian, Multinomial and Bernoulli Naive Bayes?
10. How would you select a suitable classification algorithm for a particular dataset?

Decision Trees and Ensemble Learning

1. How does a Decision Tree algorithm work?
2. What are Gini impurity and entropy in Decision Trees?
3. What is information gain, and how is it used to select a split?
4. What is pruning in Decision Trees, and why is it necessary?
5. What is the difference between a Decision Tree and a Random Forest?
6. How does Random Forest reduce overfitting compared with a single Decision Tree?
7. What is bagging, and how does it improve model performance?
8. What is boosting, and how does it differ from bagging?
9. Explain the working principles of AdaBoost, Gradient Boosting and XGBoost.
10. What is the difference between Random Forest, XGBoost and LightGBM, and when might you choose each?

Unsupervised Learning

1. What is unsupervised learning, and how does it differ from supervised learning?
2. What is clustering, and what are its real-world applications?
3. How does the K-Means clustering algorithm work?
4. What is the elbow method, and how is it used to choose the number of clusters?
5. What is the silhouette score, and what does it indicate?
6. How does hierarchical clustering work, and what is a dendrogram?

7. What is DBSCAN, and how does it identify clusters and noise?
8. What is the difference between K-Means and DBSCAN?
9. What is dimensionality reduction, and why is it useful?
10. What is Principal Component Analysis (PCA), and how does it reduce the number of features?

Model Evaluation and Performance Metrics

1. What is a confusion matrix, and how do you interpret its four outcomes?
2. What are accuracy, precision, recall and F1-score?
3. When is accuracy an unsuitable evaluation metric?
4. What is the difference between precision and recall, and when would you prioritise each?
5. What is the difference between macro, micro and weighted averaging for classification metrics?
6. What is the ROC curve, and what does the Area Under the Curve (ROC-AUC) measure?
7. What is a Precision–Recall curve, and when is it more informative than an ROC curve?
8. What are MAE, MSE, RMSE and R^2 in regression evaluation?
9. What is cross-validation, and how does K-fold cross-validation work?
10. What is stratified cross-validation, and why is it useful for imbalanced classification datasets?

Overfitting, Underfitting and Regularisation

1. What is overfitting, and what are its common causes?
2. What is underfitting, and how can you identify it?
3. Explain the bias–variance trade-off.
4. What is regularisation, and why is it used?
5. What is the difference between L1 and L2 regularisation?
6. What is the role of the regularisation parameter in a model?
7. How does cross-validation help detect overfitting?
8. How do learning curves help diagnose overfitting and underfitting?
9. What is early stopping, and how does it help prevent overfitting?
10. If a model performs well on training data but poorly on unseen data, how would you diagnose and fix the problem?

Hyperparameter Tuning and Model Selection

1. What is the difference between a model parameter and a hyperparameter?
2. What is hyperparameter tuning, and why is it important?
3. What is Grid Search, and how does `GridSearchCV` work?
4. What is Random Search, and how does it differ from Grid Search?
5. What is Bayesian optimisation for hyperparameter tuning?
6. How do you choose an appropriate evaluation metric when comparing models?
7. What is the difference between model selection and model evaluation?
8. What is nested cross-validation, and when is it useful?
9. How can hyperparameter tuning lead to overfitting on the validation dataset?
10. If two models have similar performance, how would you decide which one to deploy?

Imbalanced Data and Advanced Classification

1. What is an imbalanced dataset, and why can it be problematic?
2. What is random oversampling, and what is random undersampling?
3. What is the Synthetic Minority Over-sampling Technique (SMOTE), and how does it work?
4. Why should oversampling and undersampling be performed only within the training process during cross-validation?
5. What is class weighting, and how can it help with imbalanced classification?
6. What is threshold tuning, and how can it improve classification outcomes?
7. What is probability calibration, and why does it matter?
8. What is the difference between binary, multiclass and multilabel classification?
9. How would you build a fraud detection model when fraudulent transactions account for only 1% of the data?
10. How would you handle a classification problem in which the costs of false positives and false negatives are significantly different?

Feature Selection, Pipelines and Data Leakage

1. What is the difference between filter, wrapper and embedded feature selection methods?
2. How does Recursive Feature Elimination (RFE) work?
3. What is permutation feature importance, and how is it calculated?
4. What is the difference between feature importance in a Random Forest and permutation importance?
5. What is a Scikit-learn `Pipeline`, and why is it useful?
6. What is a `ColumnTransformer`, and how can it preprocess numerical and categorical columns differently?
7. What is target leakage, and how does it differ from ordinary data leakage?
8. How can you identify duplicate records, inconsistent labels and suspicious features before training?
9. What is the difference between random train-test splitting and time-based splitting?
10. How would you design a reproducible preprocessing and modelling workflow for a dataset containing missing values, categorical features and numerical features?

Time Series and Specialised ML

1. What is time series data, and how does it differ from cross-sectional data?
2. What are trend, seasonality, cyclic patterns and noise in time series?
3. What is stationarity, and why is it important for certain forecasting models?
4. What is the difference between AR, MA, ARMA and ARIMA models?
5. What is the purpose of differencing in time series analysis?
6. How does time series cross-validation differ from ordinary K-fold cross-validation?
7. What is the difference between point forecasting and probabilistic forecasting?
8. What is anomaly detection, and how can Isolation Forest be used for it?
9. What is recommendation-system modelling, and what is the difference between collaborative filtering and content-based filtering?
10. What is reinforcement learning, and what are an agent, environment, state, action and reward?